Systematic Review of Virtual Haptics in Surgical Simulation: A Valid Educational Tool?

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BACKGROUND: Virtual reality (VR)-based surgical simulation is an expanding and rapidly advancing modality which aims to serve the increasing demand to acquire surgical skills outside the live operating room. Haptic, or "force-feedback" technology in VR simulation is a rapidly developing field, however the role of haptics in surgical education and its efficacy is unclear.

METHODS: A systematic literature search was carried out until September 2018 in MEDLINE, Embase, and Cochrane Library using the following keywords: (VR OR VR OR simulation OR simulator) AND (Haptic feedback OR Haptics OR Force feedback) AND (Surgery). All randomized controlled studies comparing VR training with and without haptics were included. PRISMA guidelines were adhered to

RESULTS: Eight randomized controlled trials that compare VR training with and without haptics were included and 1 survey study with a total of 215 participants, 116 of which received haptic feedback and 99 were assigned to nonhaptic feedback group. Training tasks included basic proficiency based laparoscopic tasks such as object translocation, cutting, camera navigation, and more complex tasks including diathermy, suturing, dissection, knot tying, and operative maneuvers. Six randomized controlled trials demonstrated that haptic enhanced VR simulation is significantly more effective than without haptics for skill training with a reduced learning curve and faster time to proficiency and task completion, particularly in novice learners. Two studies showed no significant differences in task-assessed parameters between the haptics and nonhaptics cohorts, whereas 1 survey study suggested haptics negatively affected training with decreased realism.

CONCLUSION: Haptic feedback has been shown to improve the fidelity, realism and thus the training effect of VR simulators. However, at present haptic simulators are expensive and in a nascent stage and further research as well as cost-benefit analyses of such tools must be considered to determine whether haptics is truly a surgical necessity. (J Surg Ed 77:337−347. © 2019 Association of Program Directors in Surgery. Published by Elsevier Inc. All rights reserved.)

KEY WORDS: Haptic-feedback, Force-feedback, Laparoscopy, Simulation, Training, Surgery

COMPETENCIES: Patient Care, Medical Knowledge, Practice-Based Learning and Improvement

INTRODUCTION

Advances in surgical simulation are driven by the continuing desire to refine surgical training and optimize patient outcomes. This need to improve surgical training, in an age of surgical complexity with the advent of laparoscopic, robotic, and other minimally invasive approaches, has also been instigated by ever increasing public perception of risk and rejection of the traditional Halstedian model of apprenticeship training where the surgeon trained "on the patient."[1,2] Thus the need for realistic, efficient, and transferrable ex vivo skills training has never been greater.

Simulation training has progressed from simple bench-top knot tying boards to allowing complex full procedural training on high fidelity immersive virtual

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reality (VR) simulators. The effectiveness of simulation training, and transferability of skills from an ex vivo simulated environment to the operating room has been well established.[3–5] Furthermore, technology enhanced simulation has shown to be associated with improved performance outcomes.[3]

The utility and necessity of simulation training, and its ability to shorten learning curves, is particularly relevant given the statutory reduction of working hours during training, such as through the European Working Time Directive and analogous legislation in North America,[6] alongside the ever increasing complexity of surgical interventions.

In the current market, surgical simulators across a wide spectrum of cost, fidelity, complexity, and immersion are available.[7,8] Within general surgery, the highest fidelity VR simulators are typically applied to laparoscopic surgery. These high fidelity platforms typically tout advantages such as complex and full procedural simulation and the inclusion of haptics (or force feedback).[8] However, these technological advantages come at significant cost, with up to 50%−100% greater expense compared to lower-fidelity, nonhaptic platforms.[9] Although studies have shown low-fidelity simulators adequate to provide skill transfer to the operating room, it has been theorized that higher fidelity models with a greater degree of realism owing to haptic technology and multisensory feedback could enable richer learning experiences and outcomes.[10–13] However, the true educational value of this added functionality (and cost) is unclear.

Haptics refers to the process of recognizing objects through touch, delivered in the form of vibrations and force feedback, created by moving components of a device which is controlled by integrated software.[14] Haptic perception encompasses both tactile somatosensory perception mediated by the skin and kinaesthetic proprioceptive feedback in the tendons, muscles, and ligaments.[15] Haptic enabled systems allow the user to touch objects with a great degree of realism and is thought to be of increasing importance in fields of distance learning, interactive telemedicine, and computer simulations as it provides a bidirectional flow of information.[14,16]

Haptic technology may be used to augment VR-based simulation learning, particularly in enabling trainees to appreciate tissue structures and cultivating basic skills such as tissue tension by providing live artificial tactile feedback when excessive tension or trauma is encountered, and increasing the overall "realism" of the simulation. The effect of this on learning and skill progression, however, is not known. Laparoscopic tactile perception is often attenuated by torque exerted on laparoscopic ports (fulcrum) by the abdominal wall, particularly in obese patients. Furthermore, the lack of haptic feedback in the current generation robotic surgery platforms does not appear to have detrimentally affected outcomes when compared to laparoscopic surgery.[17–19] Conversely, it has been argued that certain haptic cues are crucial during a simulated training experience and certainly in surgery to enable appropriate and safe tissue handling.[20,21] It has been suggested that haptic feedback plays an integral role in complex motor skill acquisition; particularly the cognitive and associative stages where trainees aim to integrate task knowledge into appropriate motor behavior.[22] Thus force cues provided by haptics can allow for better task performance by allowing for a greater perception of force output − a critical element of laparoscopic skills training.

In this review, we aim to systematically assess the current evidence for the educational value of haptics in VR surgical simulation.

METHODS

Search Strategy

A systematic review was conducted in accordance with PRISMA guidelines for the reporting of systematic review. The literature search and data extraction was conducted independently by 2 authors (KR, HD) and final data including any discrepancies were resolved by consensus. A systematic search in MEDLINE, Embase, and Cochrane Library was conducted using the following keywords and MeSH headings: (Virtual Reality OR VR OR simulation OR simulator) AND (Haptic feedback OR Haptics OR Force feedback) AND (Surgery). PRISMA guidelines were adhered to in reporting the results of this study.[23]

Selection Criteria

All studies reporting data comparing outcomes for haptic feedback in VR simulation versus no haptic feedback in VR simulation were included. Studies focusing on nonhaptic feedback, projected haptic feedback models, and those studies not including a direct comparison were excluded.

No year of publication limits were set, and only English text publications were included. The search was last updated on March 1, 2019. Reference lists of included publications were also checked for any other studies suitable for inclusion. Following removal of duplicates, an initial review of titles and abstracts was conducted to identify articles of potential interest; these were then retrieved for full-text analysis and independent data extraction by authors conducting the literature search. Reference lists of retrieved articles were hand-searched for additional relevant references. Data were extracted and entered into Microsoft Excel (Microsoft Corp, Redmond, WA). A meta-analysis was not undertaken because

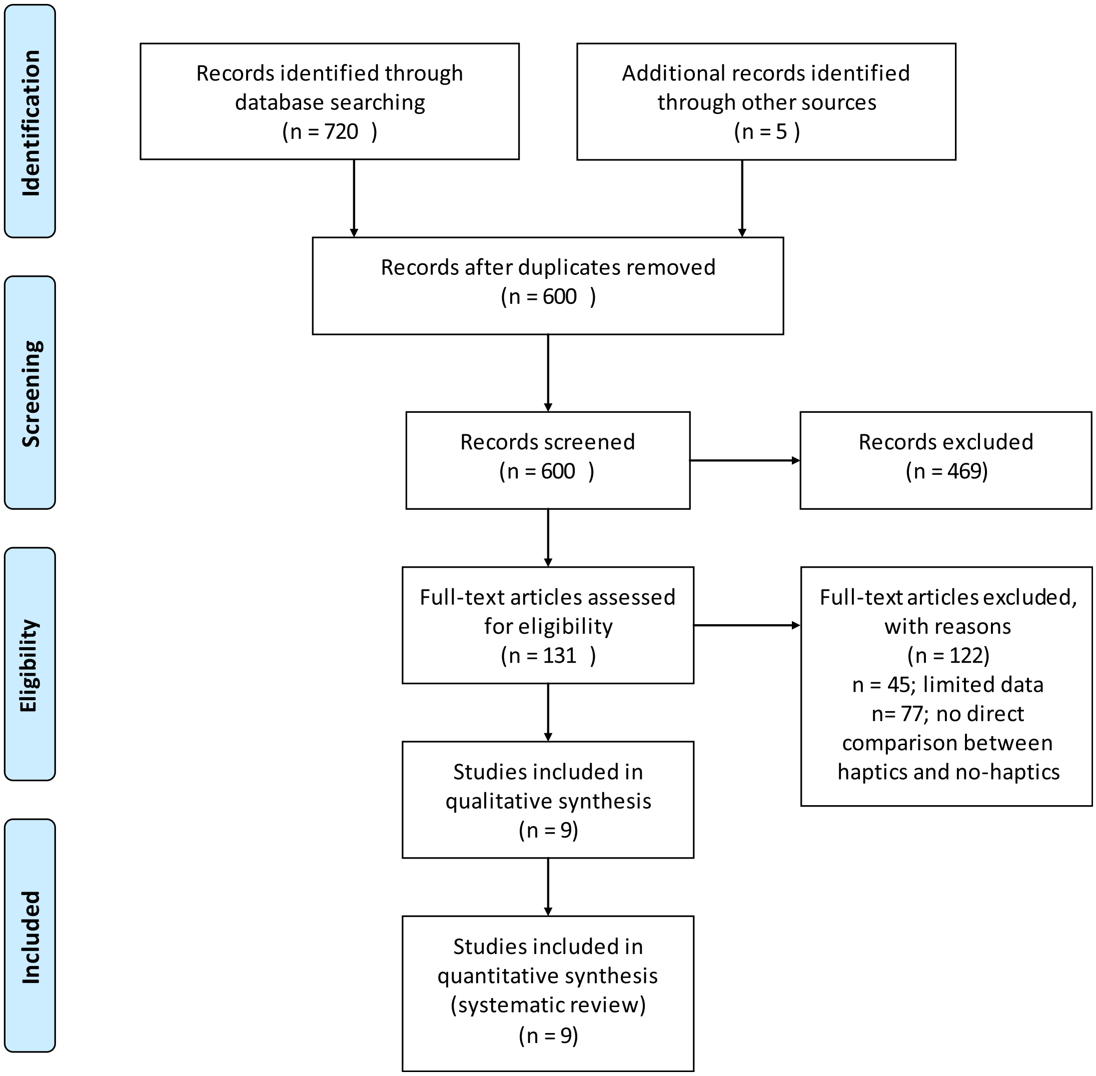

FIGURE 1. PRISMA flow diagram.

of heterogeneity of data, devices, and methods. The included studies cover a period of 13 years and multiple generations of devices. The same devices were used in no more than 2 studies; even here different metrics were used, meaning that in our judgment meaningful meta-analysis is not possible.

Results were collated and are presented in a systematic fashion. The overall findings of each study were rated by author consensus (Due to the lack of any pre-existing framework, judgment based on authors' expertize in simulation based research and is analogous to other subjective ratings whose aim is to give reader a summative overview) according to the strength of their findings with respect to the effect of haptics on training: ++ strong positive effect, + weak positive effect, +/− neutral or equivocal effect, −weak negative effect, −− strong negative effect, to provide a summative overview for the reader.

Assessment of Methodological Quality

Study quality was assessed using quality assessment tool for quantitative studies as described in the Cochrane Handbook of Systematic Reviews of interventions.[24]

RESULTS

Search Results and Demographics

Total 600 potentially relevant studies were identified in the initial search; 9 studies fulfilled the inclusion criteria and were included in final data synthesis (Fig 1). These included 8 randomized trials and 1 survey study.

Table 1 shows study characteristics including a quality assessment of each randomized controlled trial. The 9 studies (8 randomized controlled trials [RCTs] and 1

TABLE 1. Study Demographics

Author	Year	Country	Study Type	n	Participant Type	Simulation Device	Comparison	CRB Score
Kim	2004	USA	RCT	24	Novices*	Phantom 1.5	H v H2 v NH*	Medium
Strom	2006	Sweden	RCT	38	Surgeons	MIST	EH v LH	Medium
Cao	2007	USA	RCT	30	Surgeons	ProMIS/MIST	H + cognitive load v NH + cognitive load	Medium
Panait	2009	USA	RCT	10	Medical students	Laparoscopy VR	H v NH	High
Salkini	2010	USA	RCT	20	Medical students	LapMentor II	H v NH	High
Thompson	2011	USA	RCT	33	Medical students	LapMentor II	H v NH v C	Medium
Zhou	2012	USA	RCT	20	Novices**	ProMIS/MIST	H v NH	High
Vapenstad	2013	Norway	Survey	20	Surgeons	LapSim	H v NH	Medium
Hagelsteen	2017	Sweden	RCT	20	Medical students	LapSim/Simball	H+3D vision v NH	High

CRB, Cochrane Risk of Bias score; Newcastle-Ottawa Scale; RCT, Randomized Controlled Trial.
*Experience unspecified.
**Includes undergraduates and postgraduates.

survey) included a total of 215 participants, 116 of which received haptic feedback and 99 were assigned to nonhaptic feedback group. Six different simulator platforms were used in the included studies (Table 2): MIST (Mentice Medical Simulation, Sweden) (3 studies, $n = 63$), LapSim (Surgical Science, Sweden) (2 studies, $n = 30$), ProMIS (Haptica, Ireland) (2 studies, $n = 25$), LapMentor II (Simbionix corp) (2 studies, $n = 53$), Laparoscopy VR (Immersion Medical) (1 study, $n = 10$), Simball (Surgical Science, Sweden) (1 study, $n = 10$), and Phantom 1.5 (SensAble Technologies) (1 study, $n = 24$).

The haptic/nonhaptic VR training tasks varied between studies and included basic proficiency based laparoscopic tasks such as object translocation (peg transfer, object lifting), cutting, camera navigation (3 studies, $n = 64$), and more complex tasks such as use of diathermy, suturing, dissection, and knot tying (3 studies, $n = 78$). A combination of basic proficiency based tasks and complex tasks analogous to an operative maneuver was used in 3 studies ($n = 73$).

Final trainee assessment was conducted using either a simulated surgical operation (laparoscopic cholecystectomy, Heller's cardiomytomy) by 3 RCTs.[9,25,26] Two RCTs used laparoscopic knot tying as their assessment method[27,28] and 3 RCTs and the survey[20,29–31] simply used the prior training tasks used to prepare candidates for the final performance assessments.

Primary outcomes included time (6 studies), error (4 studies), path length (3 studies), economy of movement (2 studies), accuracy (2 studies), and use of cautery (2 studies). Only 1 study[25] used a global skills assessment score (OSATS) to determine performance.

Participant type varied greatly between studies with the commonest being medical students (4 studies, $n = 83$), followed by surgeons with an unspecified experience level (3 studies, $n = 88$), and finally novices (2 studies, $n = 44$). Six of these studies were conducted in the USA, 2 in Sweden and 1 in Norway.

Results of Included Studies

The included studies reported mixed results, with varying effect of haptics on outcomes. Five studies showed strong positive results[20,26–29]; these studies of haptic enabled training versus no haptics showed a positive impact of haptic feedback across multiple parameters.

Kim et al.[26] investigated the effect of 3 different levels of VR haptic fidelity on performance: the nonlinear model (most accurate high-fidelity model), the linear model (medium-fidelity), and a model without force feedback. This study using a push/pull assessment task showed that the haptic trained group with the highest fidelity took the least time for tasks. Hagelsteen et al.[28] reported that haptic feedback is a key tool for novice trainees and leads to faster acquisition of laparoscopic skills. Comparison of a range of basic laparoscopic skill including "instrument navigation", "grasping", "fine dissection", and "suturing" between a haptic and nonhaptically trained cohort revealed that the haptic trained group finished all tasks in the course faster than the group without haptic training (146 minutes compared to 215 minutes in the control group, $p < 0.002$), which was 69 minutes (32%) faster. Task complexity was assessed by Cao et al [20] who simulated this by means of increasing cognitive loading via mental arithmetic mid-simulation. In this study the combination of cognitive loading and haptic feedback showed that increasing cognitive load led to a larger increase in time-to-task completion without haptics than with haptics. Zhou et al.[27] demonstrated that haptics not only led to trainees experiencing a less steep learning curve than participants trained without haptic, but that the learning rate was also higher (70% vs 64% in No-Haptics group). They also concluded that fewer attempts

TABLE 2. Study Outcomes

Author	Training Task (s)	Assessed Task	Outcome Assessment	Outcome (data)	Findings
Kim	Seven sessions of Bimanual pushing and cutting tasks	'Basic' Heller's myotomy	Performance improvement from training	Training effect without haptics lower than with haptics ($p = 0.003$, $p = 0.81$)	++
Strom	1 hour: manipulation and Diathermy; 1 hour: point diathermy	Manipulation and diathermy in abdominal environment	Time, movement economy, tool–tool collision error, shaft–target collision error, and instrument, diathermy, and pedal error	After 2h early haptics group performed better in both tasks ($p = 0.010$, $p < 0.05$)	++
Cao	Six attempts: Object translocation	Object translocation + mental arithmetic problems	Time, error, maths problems solved	Larger increase in time-to-task completion without haptics than with haptics ($p < 0.001$) and less errors ($p < 0.001$). Haptics 37% faster, 95% more accurate	++
Panait	1 week 'warm up': Peg transfer, cutting drills	Analogous simulator drills with 3 difficulty settings	Time, tension applied, path length, errors	Time to cutting task completion shorter at all levels with haptics ($p < 0.05$)	+
Salkini	Three training tasks ×5: grasping & clipping, cutting & dissection, cautery	LC	Speed, economy of movement, accuracy	No difference in accuracy, economy of movement of dominant and nondominant hand ($p = 0.876$, 0.228, 0.6777) but improved average speed dominant hand ($p = 0.046$)	+/-
Thompson	4 h session ×2: Camera navigation, cli application, grasping, ball-drop, cutting, cautery, object translocation	LC	Time, Path length, efficiency of cautery, OSATS	No performance difference between haptics and nonhaptics	+/-
Zhou	1 h/d ×18 sessions knot tying tasks	Laparoscopic suturing and knot tying	Time, performance variation, number of knots	Learning rate 70% vs 64%, faster time to completion with haptics ($p < 0.04$), performance variance higher with no haptics ($p < 0.004$)	++
Vapenstad	fine dissection, lifting and grasping ×2 handles	Fine dissection, lifting and grasping ×2 handles	Demographics, Perception of handles	Total 95% felt friction in haptic handles was too high	--
Hagelsteen	Minimum 1 × 6h session: instrument navigation, grasping, fine dissection, suturing	Laparoscopic knot tying	Time, attempts, Path length, accuracy, tissue damage, knot error	Faster time to completion with haptics ($p = 0.002$)-69 minutes or 32% faster. Fewer attempts for proficiency with instrument navigation, grasping and suturing with haptics 27 vs 65 ($p = 0.005$), 19 vs 29 ($p = 0.017$), 24 vs 41 ($p = 0.011$)	++

Findings: ++ very strong, + strong, +/−neutral or ambiguous, −weak, −very weak.

were needed until performance plateaued. Although this study shows haptics allow for consistent and decreased variability of performance during early stages of training, the learning curves of the 2 groups converge in later sessions suggesting minimal benefits of haptics at this later stage. Finally, Strom et al.[29] compared the use of early vs late haptics. In this crossover trial, groups were exposed to either haptic or nonhaptic training followed by crossover after a

1 hour washout period. The authors demonstrated that after 2 hours of training, the early haptics group performed significantly better in both manipulation and point diathermy tasks than the group who had a later exposure to haptics ($p < 0.05$).

One study showed weakly positive results. Panait et al.[30] assessed simple and complex laparoscopic tasks with and without the use of haptics. Although individual task performances in simpler skills such as peg transfer and pattern cutting did not show any performance differences between a haptics and a nonhaptics group; the more complex cutting drill revealed a shorter time to complete tasks with haptic feedback. In addition, haptic enabled complex tasks were performed more precisely, with fewer technical errors.

Two studies reported equivocal results with no difference between a haptic trained cohort and nonhaptic trained cohort; Salkini et al.[9] and Thompson et al.[25] Both of these studies evaluated VR tasks which assessed established metrics including time, path length, speed of movement, and found no significant differences between a haptic feedback enabled cohort and a nonhaptic cohort. However it was noted that in the Salkini study[9] across both cohorts the video-game playing demographic had more economic and faster hand movement than their colleagues. Authors reported a trend toward improved outcomes in terms of economy of both hand movements and speed of nondominant hand, although statistically not significant. However the average speed of the dominant hand was faster with haptics enabled.

One study showed strongly negative results. Vapenstad et al.[31] reported a survey study of a haptic enabled device vs a nonhaptic enabled device. Surgeons with varying laparoscopic experience levels were asked to give an opinion on 2 instrument ports commonly used in VR simulators: the Xitact IHP (instrument haptic port) with haptic feedback and the Xitact ITP (instrument tracking port) without haptic feedback. These instruments were connected to the LapSim VR simulator. The majority of participants preferred the simulated device without haptics. Although 15 of 19 surgeons (79 %) claimed handles with haptic feedback were important to simulate realism, only 4 (20 %) of the participants said that the handles with haptic feedback imitated reality best whilst 14 surgeons (70 %) thought the handle without haptic feedback felt most realistic.

Assessment of Methodological Quality

Across the 8 RCTs included, 7 out of 8 studies contained 3 or more domains showing evidence of limitations as defined by the Cochrane Risk of Bias Tool, suggesting the risk of bias for these studies was moderate to high. The summary of each study's risks of biases are

TABLE 3. Risk of Bias Summary

Author	Selection Bias Random Sequence Generator	Selection Bias Allocation Concealment	Performance Bias Blinding of Participants	Detection Bias Blinding of Outcome Assessors	Attrition Bias Incomplete Collection/ Outcome Data	Reporting Bias Selective Reporting	Other Biases Baseline Confounders
Kim	No	No	No	?	Yes	Yes	Yes
Strom	?	Yes	No	No	Yes	Yes	Yes
Cao	No	No	No	No	Ys	Yes	Yes
Panait	No	No	No	N	Yes	Yes	Yes
Salkini	No	No	No	No	Yes	Yes	No
Thompson	Yes	Yes	No	No	Yes	No	Yes
Zhou	No	No	No	No	Yes	No	Yes
Vapenstad	No	Yes	Yes	?	Yes	No	Yes
Hagelsteen	Yes	No	No	No	No	No	yes

Yes, component present and adequate; no, Component absent or inadequate; ?, Complete unclear/data unavailable.

summarized in Table 3. Table 1 shows the risk of bias summary across studies.

DISCUSSION

This review highlights the inconsistency and paucity of current evidence with regards to haptics and its validity and evidence of its value in training. Haptic feedback as a concept is an important factor in minimally invasive surgery (MIS) and although its role is contradictory, this has not limited the development of haptic enabled technologies.

Overall the results suggests that there is a role for haptic feedback in surgical simulation training with 6 studies[20,26–30] demonstrating a positive effect of training performance when compared to no haptic feedback on a range of tasks from fundamental laparoscopic skills to full simulated operations. These 6 studies show improved performances across multiple validated assessment parameters with haptics as opposed to no haptics. Studies which included an assessment of participants learning curves[27,28] demonstrated that haptic feedback enhanced learning, resulting in an upward trend in terms of task proficiency compared with those trained without haptic feedback; these results were particularly significant at the early stages of training.

Although there has been previous validation of skill transfer from VR simulators that do not incorporate haptic feedback to the operating room,[32] there remains controversy regarding the timing of haptics integration into surgical training. Panait et al.[30] in their study of laparoscopic novices performing simple tasks with and without haptic demonstrated improved performance with haptic feedback, strengthening the case for early haptic induction. These findings were congruent with Zhou[27] who found that complex tasks were learned more efficiently when haptic feedback was present during the early phases of surgical training and identified this early phase as the key point to introduce and emphasize haptic cues.

The application of, and need for haptics may vary with the complexity of task at hand. Chmarra et al.[33] suggest that procedures requiring force application, such as laparoscopic knot tying, or handling the bowel would require simulators with haptic feedback; whereas simpler tasks such as peg transfer or tasks relying heavily on hand-eye coordination as is the case with the tasks assessed in the 3 studies above, can be trained on simulators with no haptic feedback.[9,25,31] Conversely, at higher levels of complexity, current simulator limitations in graphic and physical tissue modeling may reduce the usefulness of VR training for complex full procedure simulation both with or without haptics; this may be expected to change as software models continue to rapidly evolve.

Highlighting the point regarding the relationship between task complexity and utility of haptics, Hagelsteen[28] shows that there was a lack of statistically significant differences in performance between the 2 study cohorts in a "fine dissection" task; a fairly basic task. The authors concluded that the simplicity of said task may have resulted in a ceiling effect; suggesting a threshold level of task complexity is required to affect tested outcomes. It would be of interest to further investigate this ceiling effect and the thresholds of task complexity required; especially in the light of similar basic tasks in the Thompson study[25] suggesting haptics had no effect on performance. Indeed Cao et al.[20] who explored the effect of haptic feedback on both surgical experience and cognitive load by adding a secondary less demanding task showed that these "complex" tasks (due to the additional mental load) were performed better with haptic feedback and these participants demonstrated reduced cognitive load. Of particular interest were the subgroup analyses of experienced surgeons in the no-cognitive-load subgroup who displayed the largest performance improvement thus implying that experts use spare cognitive capacity to address and use indirect haptic cues.

It is well known that during MIS, tactile feedback is attenuated and distorted due to interactions of the instruments and ports with the abdominal wall but these physical cues are often important in the surgical setting to encourage safe tissue handling skills.[34]

Studies have suggested previously that the incorporation of haptic feedback enabled instruments, are required to increase efficiency in terms of more successful grasping actions and to provide greater control over tissues.[35] The results of 2 recent reviews regarding haptic feedback in MIS conclude that the implementation of haptic technology in MIS may prove beneficial to both surgeon and patient.[32,36]

The future of haptic integration in surgical simulation includes application of this in robotic assisted surgery and further technological design improvements—perhaps a future goal is to develop surgical instruments which serve as extensions of the surgeon's hands and allows for measurement of contact forces between the device and tissue.[37] While current robotic platforms do not offer integrated tactile feedback, it has been hypothesized that the inclusion of haptic feedback in robotic surgery may reduce technical errors, tissue damage, and decrease operative time.[34] A preliminary study by Jacobs et al. demonstrated the potential value and impact of robotic training with a combination of haptic and visual feedback. The results indicated that haptic feedback improved training performances with respect to task completion time, accuracy, and number of errors made compared with task performances with only visual feedback.[38]

Despite the perceived benefits of haptic technology, it remains in a nascent stage Haptic feedback could be improved in several ways; improving the actuators which provide the haptic cues and the use of more realistic tissue models to derive more natural responses and interactions with tissue and structures.[39]

Study Limitations

The studies assessed in this review are subject to a number of limitations. A number of different simulators were used with varying levels of validation of these instruments, limiting the interpretation of trial results and scope of successful transfer of skills to the OR environment. Thus, inconsistencies in both type of simulators used, the method of training sessions and performance assessments as well as the disparity across methodology and study outcomes limits generalizability of the results. In light of the above, a meta-analysis of studies was deemed inappropriate given the wide range of simulators and metrics.

With regards to bias and evidence quality, the results summarized in Table 3 suggest overall low quality of evidence with a medium/high degree of bias across most studies. Limitations endemic to most of these studies which limits quality and introduces bias include the use of small cohorts and novices participants and volunteers; the latter resulting in self-selection and selection bias.

Of the studies included, only 1 study had any form of validated structures assessment of laparoscopic surgery or skills assessment (GOALS/OSATS).[40] Whilst all studies attempted to compare groups using performance metrics provided by the simulator devices, this raises the question as to which variables and metrics provide the best measurement of skill acquisition across the different platforms, as different simulators use different metrics for objective measurement of skills acquisition. This lack of a consistent objective scoring system makes a direct comparison of results after training on different simulators difficult.

While the majority of studies reported positive effects of haptics, there was 1 negative study in this review.[31] Considering this study, however, the majority of participants suggest a lack of realism with the haptic handset itself (rather than haptics as a concept or tool) and attributed this to increased stiffness. This may in fact be a comment on an inferior product, rather than haptics as a concept, particularly as no objective learning outcomes are reported.

Further work is needed to improve the provided quality or even type of haptic feedback. A review by Coles et al.[8] suggested that the right amount of haptic feedback is needed at the right time and that providing feedback of too high fidelity can be as much a hindrance as providing too little haptic feedback. This is particularly evident in laparoscopic surgery, where instruments entering through tight ports are exposed to frictional forces that mask the interactive forces at the instrument tip. Thus providing clear interaction forces in training environs may negatively affect the real intraoperative training experience as the user would not be fed back this level of quality feedback.

Finally, the rapid progression of computer-based software and hardware, and generational advance of simulator technology, also means that an objective review of published literature risks being out of date with reference to currently available platforms. Many of these already offer new advances on realism (such as immersive headset-based VR), and updated software and haptic sets, but may not yet be reflected in published academic literature.

CONCLUSION

Although a vastly improved domain, the development and state of tactile simulators is still in its infancy and will have an increasingly pivotal role to play in the medical domain as technology grows. Despite demonstrating its efficacy for training in minimal access surgery or where an instrument acts as an indirect tactile conduit (needles, drills); the field of open surgery and procedures requiring direct physical contact with soft tissues directly will remain a challenge to simulate.

With an ever diminishing number of training hours available for trainees and the increasing complexity of procedures; simulation plugs a vital gap in training. It is clear that with appropriate use and direction, training that is haptic-enabled is a key adjunct to training, proficiency, and improving patient outcomes.

Despite the high cost of these tactile VR simulators in comparison to lower cost haptic-free models, few studies have examined the added value of this feature. The cost-benefit analysis of such a profitable marketing tool must be considered and whether haptics is truly a surgical necessity. Haptics as part of a move toward "ultra-realistic" simulators have proven their use in other fields such as flight simulation training and space exploration to good effect; further research is required to assess the need for universal inclusion of haptics in surgical simulation.

ETHICAL APPROVAL

Ethical approval was not required for this study.

AUTHOR CONTRIBUTION

Conception and design: PP, KR, HD.
Analysis and interpretation: KR, PP, HD.
Data collection: KR, HD, PP.
Writing the article: KR, PP, HD.
Critical revision of the article: KR, PP, HD.
Final approval of the article: KR, PP, HD.
Statistical analysis: KR, PP, HD.
Overall responsibility: KR, PP, HD.
(KR – Karan Rangarajan, PP – Philip Pucher, HD – Heather Davis).

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